

Software Requirements Specification (SRS)

Student Services Management System (SSMS)

1. Introduction

1.1 Purpose

The Student Services Management System (SSMS) is a web and mobile-based application designed to streamline communication and academic interactions between students, faculty, and administrators. The system provides a centralized platform for:

- Student Notices
- Leave Applications
- Grievance Management
- Assignment Submission

The objective is to eliminate paperwork while ensuring transparency, timely communication, and efficient management of student-related services.

1.2 Scope

The system will provide:

- Institute-wide notices
 - Department/Class-specific notices
 - Online leave application and approval
 - Student grievance registration and tracking
 - Assignment upload and evaluation
 - Email/App notifications
 - Dashboard for students, faculty, and administrators
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1.3 Users

User	Responsibilities
Student	View notices, submit leave, grievances, assignments

User	Responsibilities
Faculty	Approve leave, review grievances, manage assignments
HOD	Monitor department activities
Administrator	Manage users, notices, departments, reports

2. Overall Description

2.1 Product Perspective

The system will consist of:

- Student Mobile App
 - Faculty Portal
 - Admin Portal
 - Backend API
 - Database
 - Notification Service
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2.2 Product Features

Student Module

- Login
 - Dashboard
 - View Notices
 - Apply Leave
 - Raise Grievance
 - Submit Assignments
 - Track Status
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Faculty Module

- Login
- Dashboard
- Publish Assignments
- Review Assignments
- Approve Leave
- Resolve Grievances

- Publish Notices
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HOD Module

- Department Dashboard
 - Monitor Faculty
 - View Reports
 - Resolve Escalated Grievances
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Admin Module

- User Management
 - Department Management
 - Notice Management
 - Report Generation
 - System Configuration
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3. Functional Requirements

Module 1: Student Notice Management

Description

Allows institute administration and faculty to publish notices.

Functional Requirements

ID	Requirement
FR-001	Admin can create notices
FR-002	Faculty can create class notices
FR-003	Student can view notices
FR-004	Student receives notification
FR-005	Search notices
FR-006	Filter notices by department

ID	Requirement
FR-007	Archive old notices

Notice Fields

- Notice ID
 - Title
 - Description
 - Department
 - Semester
 - Attachment
 - Publish Date
 - Expiry Date
 - Priority
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Module 2: Leave Management

Description

Students can apply for leave digitally.

Functional Requirements

ID	Requirement
FR-101	Student submits leave application
FR-102	Upload supporting document
FR-103	Faculty reviews leave
FR-104	Faculty approves/rejects
FR-105	Student receives notification
FR-106	Leave history available

Leave Form

- Student Name
- PRN
- Department
- Semester
- From Date

- To Date
- Reason
- Attachment

Leave Workflow

Student



Submit Leave



Faculty



Approve / Reject



Notification



Student Dashboard

Module 3: Grievance Management

Description

Students can register complaints.

Categories

- Academic
- Examination
- Hostel
- Library

- Fees
 - Discipline
 - Transport
 - Others
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Functional Requirements

ID	Requirement
FR-201	Student submits grievance
FR-202	Upload evidence
FR-203	Faculty/HOD assigns grievance
FR-204	Track grievance status
FR-205	Add remarks
FR-206	Close grievance
FR-207	Student provides feedback

Grievance Status

- Submitted
 - Under Review
 - Assigned
 - In Progress
 - Resolved
 - Closed
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Module 4: Assignment Submission

Description

Faculty uploads assignments while students submit solutions.

Faculty

- Create Assignment
- Upload PDF

- Set Deadline
 - Set Marks
 - Publish Assignment
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Student

- View Assignment
 - Download Assignment
 - Upload Solution
 - Resubmit (before deadline)
 - View Marks
 - Faculty Feedback
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Functional Requirements

ID	Requirement
FR-301	Faculty creates assignment
FR-302	Student views assignment
FR-303	Student uploads assignment
FR-304	Deadline validation
FR-305	Faculty downloads submission
FR-306	Faculty evaluates
FR-307	Student views marks

Assignment Details

- Assignment ID
 - Subject
 - Title
 - Description
 - Faculty
 - Deadline
 - Maximum Marks
 - Attachment
-

4. User Roles

Student

- Login
 - View Notice
 - Apply Leave
 - Raise Grievance
 - Submit Assignment
 - Track Status
-

Faculty

- Login
 - Publish Notice
 - Approve Leave
 - Review Assignment
 - Resolve Grievance
-

HOD

- Monitor Department
 - Reports
 - Escalation Handling
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Administrator

- Manage Users
 - Manage Departments
 - Manage Subjects
 - Reports
 - Notifications
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5. Non-Functional Requirements

Performance

- Response time < 3 seconds
 - Support 5,000+ users
 - 99.9% uptime
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Security

- Role-Based Access Control (RBAC)
 - Secure Login
 - Password Encryption
 - HTTPS Communication
 - JWT Authentication
 - Audit Logs
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Reliability

- Automatic Backup
 - Disaster Recovery
 - Data Integrity
 - High Availability
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Usability

- Mobile Friendly
 - Responsive UI
 - Simple Dashboard
 - Accessibility Support
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6. Database Design

Student

- StudentID
- PRN
- Name
- Email

- Department
 - Semester
 - Mobile
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Faculty

- FacultyID
 - Name
 - Department
 - Subject
 - Email
-

Notice

- NoticeID
 - Title
 - Description
 - Attachment
 - PublishDate
 - ExpiryDate
-

Leave

- LeaveID
 - StudentID
 - FromDate
 - ToDate
 - Reason
 - Status
-

Grievance

- GrievanceID
- StudentID
- Category
- Description

- Status
 - Remarks
-

Assignment

- AssignmentID
 - Subject
 - FacultyID
 - Deadline
 - Marks
-

Submission

- SubmissionID
 - AssignmentID
 - StudentID
 - FilePath
 - SubmissionTime
 - Marks
 - Feedback
-

7. Notifications

The system shall notify users for:

- New Notice Published
- Leave Approved/Rejected
- New Assignment Published
- Assignment Deadline Reminder
- Assignment Evaluated
- New Grievance Update
- Grievance Resolved

Notifications shall be delivered via:

- In-App Notifications
- Email
- Push Notifications (Mobile App)

8. Reports

Administrator Reports

- Student Activity Report
- Leave Report
- Grievance Report
- Assignment Submission Report
- Department-wise Statistics
- Faculty Performance Report

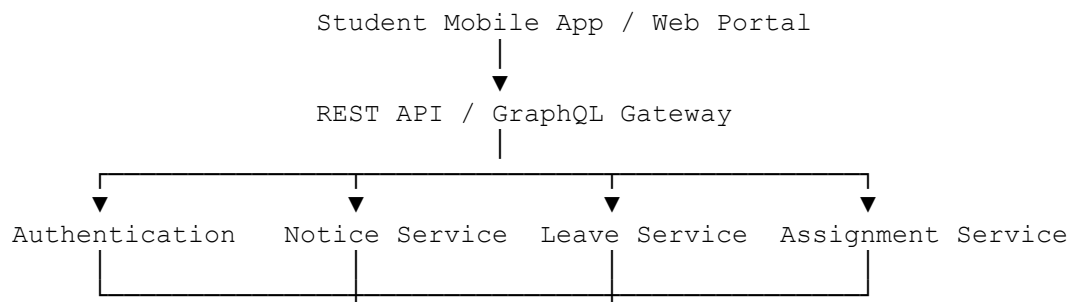
Faculty Reports

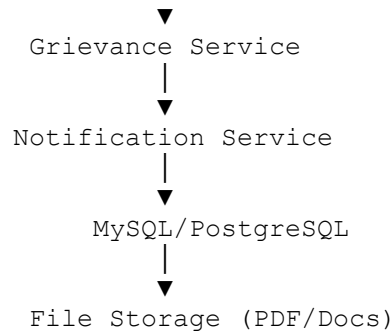
- Assignment Submission Status
- Leave Requests
- Grievance Status
- Student Participation

Student Reports

- Leave History
 - Submitted Assignments
 - Grievance History
 - Notice Archive
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9. System Architecture





10. Technology Stack (Suggested)

Layer	Technology
Frontend	React.js / Angular, Flutter (Mobile)
Backend	Spring Boot / Node.js (Express)
API	REST API or GraphQL
Database	PostgreSQL / MySQL
Authentication	JWT, OAuth 2.0
File Storage	AWS S3 / Azure Blob / Local Storage
Notifications	Firebase Cloud Messaging (FCM), Email (SMTP)
Deployment	Docker, Kubernetes, Nginx
Cloud	AWS, Azure, Google Cloud

11. Future Enhancements

- Attendance Management
- Timetable
- Internal Marks
- Online Fee Payment
- Event Registration
- Placement Module
- Library Integration
- Student Feedback System
- AI-powered Chatbot for Student Queries
- ERP Integration
- Single Sign-On (SSO) with Institute Credentials

This SRS provides a comprehensive blueprint for developing a **Student Services Management System (SSMS)** focused on **Notices, Leave Management, Grievance Handling, and**

Assignment Submission, and is suitable for implementation as a web and mobile application within an engineering institute.